Methodology Number	FO-001 Ver.3.2
Methodology Name	Forest management activity

<Subject of Methodology>

- This methodology applies to activities to secure removals through the implementation of forest management or protection.

<Definition of Terms>

Term	Definition
Forests for which project plan is registered (see eligibility criterion 1)	The whole or part of a forest where a forest management plan has been established and forest management practices are carried out in accordance with the forest management plan, within the scope of obligations specified in ① of 8.1.2 of Implementation Rule (for project participants)
Project site	Forests for which removals and emissions are to be calculated. Forests subject to calculation of removals are the forest stands for which the project plan is registered that: ① is a forest stand of an artificial forest where forest management (reforestation, nurturing, thinning) was carried out between April 1990 and the start of the certification period and where forest protection is implemented during the certification period, ② is a forest stand where forest management (reforestation, nurturing, thinning) was carried out on or after the start date of the certification period (the applicable forest stand is selected arbitrarily), and ③ is a forest stand of a natural regenerated forest where forest protection is implemented (Only areas designated as restricted forests, such as forest reserves, etc.) (Forest stands that fall under ①② and ③ are arbitrarily sampled). The target for calculating emissions is the forest stands for which the project plan is registered where final cutting is to be conducted within the certification period (All applicable forest stands are sampled).
Forest management	Afforestation (planting, ground preparation, picking), nurturing (weeding, climber cutting, clean cutting, pruning), thinning, and final cutting
Forest protection	Extermination and prevention of forests pests, prevention of bird and animal damage, prevention of fire, confirmation of boundaries, and forest patrols
Harvested timber	Timber (unprocessed) that has been cut down and is to be shipped for use in lumber, plywood, or as a raw material (wood chips to be used in pulp, wood boards, and fuel, etc.). Timber for other materials (sleepers, utility poles, stakes, scaffolding logs, etc.) and shiitake production are not included in harvested timber.

1. Eligibility Criteria

This methodology may be applied if all of the following criteria are satisfied.

- Criterion 1: The project plan is registered for the entire area of the forest management plan that has been approved by the mayor of the municipality, and forest management is carried out in accordance with the forest management plan.

  However, in cases where it is difficult to register a project plan for the entire area of the forest management plan, such as when a forest management plan is established by multiple forest owners, the project plan may be registered only for the forest in the forest management plan where the project is to be implemented that is owned or managed by the project participant itself (its entire area).

  In addition, if all of the following requirements are satisfied, a project plan may be registered by selecting a portion of the forests owned or managed by the project participant itself.

  - Requirement 1: The are is 500 ha or greater.
  - Requirement 2: It is recognized that the location has not been arbitrarily selected, such as through the deliberate exclusion of final cutting sites.
- Criterion 2: When a project site determined based on criterion 1 includes stands where final cutting is carried out, the cumulative amount of expected removals during the certification period is positive.
- Criterion 3: The project has at least one area in the project site where thinning within the certification period is planned based on the forest management plan. However, if the project site cannot include forests subject to planned thinning, afforestation or nurturing based on the forest management plan is planned at one or more locations in the project site.
- Criterion 4: No land conversion of the project site (except for unavoidable land conversions such as expropriation) is planned in the forest management plan.
- Criterion 5: The forest management plan is continuously prepared during the certification period and up to the date when 10 years have elapsed from the end of the certification period, and it is understood that if certification of the forest management plan is not continued, duty of replenishment occurs in accordance with 8.1.3 of Implementation Rule (For project participants).

<Explanation of the eligibility criteria>

If multiple forest management plans are combined and project application made, the applicable individual forest management plans must satisfy the eligibility criteria. However, regarding criterion 2, the project application as a whole, rather than individual forest management plans, may meet the eligibility criteria.

If the remaining plan period of the forest management plan in effect at the time of the application for validation is less than one year, the application shall include the scheduled plan for the next forest management plan and validation be received.

Criterion2:

"Owned or managed by the project participant itself" refers to the scope of ownership if the project participant itself is the owner of the forest; or if the project participant is the manager under a contract for forest management, it refers to the scope of the contract with the consignor. If, in accordance with a proviso, the applicant intends to implement a project by selecting only forests owned or managed by the applicant itself, a

copy of the entire forest management plan must be submitted at the time of application for project registration.

"Arbitrarily selected" as stipulated in Requirement 2 is where the project participant registers a project plan by selecting a portion of the forest owned or managed by the project participant itself refers to cases in which a portion of a forested area is selected, for example, by excluding ridges only along valley lines with fast growth, by unnaturally traversing a watershed while ignoring topography such as ridges or valley lines, or by intentionally reducing or excluding final cutting areas. "Non-arbitrarily selected" is where a geographically coherent area, such as a city, town or village-based unit, is selected.

If changing the project site once registered after validation, validation is required again to confirm that the reason for the change is unavoidable and that the new project site meets all the requirements.

Criterion2:

In proving that the cumulative total of expected removals during the certification period is positive, a simplified method, such as setting a conservative site index may be used if there is no actual data, etc. Information such as forest registry, logging notice, or forest management plan may also be used, to the extent that the validation authority can be assured of this.

Even if the criteria is met, for projects that temporarily have negative cumulative expected removals depending on the timing of final cutting, requests for certification of credit may only be made after this period has passed.

In calculation of expected removals, if reforestation after final cutting is planned and the project chooses to account for the amount of carbon stock (that converted to tons of CO2, hereinafter the same applies) by an equivalent selected based on the provisions of "3. Calculation of removals after project implementation" of this methodology, whereby the standard cutting age of the forest stand in the reforested area after final cutting (this means the lowest forest age for final cutting as stipulated in Article 38(v), Article 39 (1) and Article 39(2)(ii) of Ordinances of Enforcement of the Forest Act (Order of Ministry of Agriculture and Forestry No. 54 of 1951); hereinafter the same applies), the removals may also be included in the expected removals as standards for certifying a forest management plan In addition, if final cutting and removal/thinning are planned, the removals related to harvested timber, which is calculated based on "3. Calculation of removals after project implementation" of this methodology, can be included in the expected removals.

Criterion3:

"Forests subject to planned thinning" refers to those specified in Article 38 (iii) of the Ordinances of Enforcement of the Forest Act.

Criterion2:

Documentation indicating the intent to continue the forest management plan and understanding of duty of replenishment must be submitted at the time of the application for validation.

2.Calculation of removals

Removals in this methodology shall be calculated for each year during the certification period.

(Formula 1)

Symbol	Definition	Unit
C_{total}	Removals in the fiscal year	tCO2
C_{PJ}	Removals after project implementation in the fiscal year	tCO2
C_{cut}	Removals after project implementation in the fiscal year	tCO2
C_{BL}	Baseline removals for the relevant fiscal year	tCO2

<Greenhouse gas emissions/removals activities to be taken into account in the calculation of removals>

Item	Emissions activities Removal activities	Greenhouse gas	Description
Baseline Removals	Increase in above- and below-ground biomass	CO_2	Removals if reforestation is not implemented
After the project implementation Removals	Increase in above-ground biomass	CO_2	Removals due to increase in above-ground biomass as a result of forest management activities
	Increase in below-ground biomass	CO_2	Removals due to increase in below-ground biomass as a result of forest management activities
	Increase in carbon fixation of harvested timber	CO_2	Removals related to carbon fixation for harvested timber removed as a result of forest management activities (final cutting and thinning) that is used over a period of time that is considered permanent
After the project implementation emissions	Decrease in above- and below-ground biomass	CO_2	Emissions due to reduction of above- and below-ground biomass due forest management activities (final cutting

3. Calculation of removals after project implementation

$$C_{PJ} = C_{PJ,AG} + C_{PJ,BG} + C_{PJ,RF,AG} + C_{PJ,RF,BG} + C_{PJ,WP} \quad \text{(Formula 2)}$$

Symbol	Definition	Unit
C_{PJ}	Removals after project implementation in the fiscal year	tCO2
$C_{PJ,AG}$	Removals by above ground biomass in the fiscal year	tCO2
$C_{PJ,BG}$	Removals by below-ground biomass in the fiscal year	tCO2

$C_{PJ,RF,AG}$	Removals by above-ground biomass until the stand reforested in the relevant fiscal year reaches the standard cutting age, etc.	tCO2
$C_{PJ,RF,BG}$	Removals by below-ground biomass until the stand reforested in the relevant fiscal year reaches the standard cutting age, etc.	tCO2
$C_{PJ,WP}$	Removals related to the carbon fixation of harvested timber for the relevant fiscal year that is to be used for a period of time considered to be permanent	tCO2

<Supplementary explanation>

- $C_{PJ,RF,AG}$ and $C_{PJ,RF,BG}$ shall be included in the calculation only when final cutting has been carried out during the certification period and when artificial afforestation (reforestation) has been carried out at the final cutting site in accordance with the forest management plan (only when the renewal after harvesting is planned as plantation forestation in the forest management plan in effect at the time of final cutting). The removals shall be limited to an amount equal to the emissions ($C_{cut,AG}$ and $C_{cut,BG}$) associated with the corresponding final cutting, respectively.
- The calculation of $C_{PJ,RF,AG}$ and $C_{PJ,RF,BG}$ is optional (if not included in the calculation, both should not be included in the calculation), and if calculated, $C_{PJ,AG}$ and $C_{PJ,BG}$ cannot be calculated for the reforested stand until the forest age of the stand reaches the standard cutting age, etc., and if $C_{PJ,RF,AG}$ and $C_{PJ,RF,BG}$ are not calculated, $C_{PJ,AG}$ and $C_{PJ,BG}$ can be calculated for the reforested stand.
- The calculation of $C_{PJ,WP}$ is optional and can be omitted from the calculation.

a) Removals by above-ground biomass (excluding stands for which $C_{PJ,RF,AG}$ is calculated)

（Formula 3）

Symbol	Definition	Unit
$C_{PJ,AG}$	Removals by above-ground biomass for the relevant fiscal year	tCO2
$C_{PJ,AG,i}$	Removals by above-ground biomass during the relevant fiscal year in the artificial forests in class i where forest management (afforestation, nurturing, and thinning) was carried out since April 1990, and removals by above-ground biomass during the relevant fiscal year in natural regenerated forests in class i where forest protection was implemented from the start of the certification period (only areas designated as restricted forests, such as forest reserves, etc.).	tCO2
$Area_{Forest,i}$	Of the artificial forests in class i where forest management was carried out from April 1990 to the relevant fiscal year, the area of stands where forest management (excluding final cutting) or forest protection has been implemented from the start of the certification period until the relevant fiscal year (actual measured area multiplied by 0.9), and area of natural regenerated forests in class i where forest protection was implemented (only areas	ha

	designated as restricted forests, such as forest reserves, etc.) from the start of the certification period until the relevant fiscal year (Calculated by multiplying the measured area by 0.9 for stands where forest pest extermination and prevention activities have been conducted. Area is that in the forest registry for forest areas where fire prevention and other protection activities have been implemented.)	
$Trunk_{SC,i}$	Annual stem volume growth per unit area for the relevant fiscal year in class i	m^3/ha
WD_i	Factor (volume density) for converting stem volume growth in class i to biomass (dry weight)	t/m^3
BEF_i	Factor (expansion factor) for adding branch and leaf biomass to stem biomass in class i	—
CF	Factor for converting biomass amount (dry weight) to carbon amount (Carbon content)	—
i	Class of the forest by forest type (artificial forest, natural regenerated forest), tree species, forest age, site index, etc.	—

Supplementary explanation

- $Area_{Forest,i}$ of artificial forest is the area of forest stands classified as class i (class according to forest type, species, forest age, site index, etc.) where the implementation of forest management (reforestation, nurturing, thinning and final cutting) from April 1990 to the relevant fiscal year when the removals are calculated can be proved by forest management records, logging notices, and grant related documentation, and photos that can be used to determine traces and timings of previous work in the relevant forest stand, and for which appropriate forest management (excluding final cutting) or forest protection have been implemented in accordance with the forest management plan from the start of the certification period to the relevant fiscal year. Nurturing is weeding, climber cutting, clean cutting, and pruning. Forest protection includes pest and animal control, fire prevention, confirmation of boundaries, and forest patrols, and the condition of the forest is checked against the description in the PDD at the time of implementation.

  The area shall be calculated by multiplying by 0.9 the actual measured area where forest management was implemented from April 1990 to the relevant fiscal year.
- $Area_{Forest,i}$ of natural regenerated forest is the area of forest that is restricted forests (forests designated as forest reserves, regions where preservation is implemented, national parks (limited to special protection zones, class 1 special zones, and class 2 special zones), quasi-national parks (limited to special protection zones, class 1 special zones, and class 2 special zones), nature conservation area special zones, and special seed tree forests) and classified as class i (class according to forest type, species, forest age, etc.) where forest protection has been implemented based on the forest management plan from the start of the certification period until the relevant fiscal year. The area of forest protection in which extermination and prevention of forest pests is carried out shall be the area of the measured area where tree species subject to

protection activities are standing multiplied by 0.9, and the area where fire prevention and other forest protection is conducted shall be the area in the forest registry of all sub-compartments whose forest type is naturally regenerated forest included in the forest compartment in which the activity was carried out.

- Forest protection in artificial forests and natural regenerated forests that are $Area_{Forest,i}$ includes patrol by aircraft ("aircraft" as defined in Article 2 (1) of the Civil Aeronautics Act (Act No. 231 of 1952) and "unmanned aircraft" as defined in (22) of the same article).

b) Removals by below-ground biomass (excluding stands for which $C_{PJ,RF,BG}$ are calculated)

(Formula 4)

Symbol	Definition	Unit
$C_{PJ,BG}$	Removals by below-ground biomass in the fiscal year	tCO2
$C_{PJ,BG,i}$	Removals by below-ground biomass for the relevant fiscal year in the artificial forests in class i where forest management (afforestation, nurturing, and thinning) has been carried out since April 1990, and removals by below-ground biomass during the relevant fiscal year in naturally regenerated forests in class i where forest protection was implemented from the start of the certification period (only areas designated as restricted forests, such as forest reserves, etc.).	tCO2
$C_{PJ.AG,i}$	Removals by above-ground biomass during the relevant fiscal year in the artificial forests in class i where forest management (afforestation, nurturing, and thinning) was carried out since April 1990, and removals by above-ground biomass during the relevant fiscal year in natural regenerated forests in class i where forest protection was implemented from the start of the certification period (only areas designated as restricted forests, such as forest reserves, etc.).	tCO2
$R_{ratio,i}$	Factor (underground ratio) for adding removals of the below-ground part (roots) to the removals of the above-ground biomass in class i	—
i	Class of the forest by forest type (artificial forest, natural regenerated forest), tree species, forest age, site index, etc.	—

c) Removals by above-ground biomass until the stand reforested after final cutting reaches the standard cutting age, etc.

$$C_{PJ,RF,AG} = \sum_i C_{PJ,RF,AG,i} = \sum_i (Area_{Forest,RF,i} \times Trunk_{SC,RF,i} \times WD_i \times BEF_i \times CF \times \frac{44}{12}) \quad (式\ 5)$$

Symbol	Definition	Unit
$C_{PJ,RF,AG}$	Removals by above-ground biomass until the stand reforested in the relevant fiscal year reaches the standard cutting age, etc.	tCO2
$C_{PJ,RF,AG,i}$	Removals by above-ground biomass until the class i stand reforested in the relevant fiscal year reaches the standard cutting age, etc.	tCO2

$Area_{Forest,RF,i}$	Area of the forest stand classified as class i where forest management (reforestation) is implemented in the relevant fiscal year after final cutting	ha
$Trunk_{SC,RF,i}$	Stem volume per unit of area in reforested class i stand standard cutting age, etc.	m^3/ha
WD_i	Factor (volume density) for converting stem volume growth in class i to biomass (dry weight)	t/m^3
BEF_i	Factor (expansion factor) for adding branch and leaf biomass to stem biomass in class i	—
CF	Factor for converting biomass amount (dry weight) to carbon amount (Carbon content)	—
i	Class of the forest by forest type (artificial forest, natural regenerated forest), tree species, forest age, site index, etc.	—

Supplementary explanation

- $C_{PJ,RF,AG,i}$ shall be limited to an amount equal to the above-ground biomass emissions ($C_{CUT,AG}$ calculated in Formula 9) from the final cutting implemented prior to reforestation (during the certification period) at the relevant reforestation location (if final cutting took place at the relevant forest stand during the certification period and $C_{CUT,AG}$ was not calculated, $C_{PJ,RF,AG,i}$ also cannot be calculated).
- The standard cutting age, etc. referred to in $Trunk_{SC,RF,i}$ means the lower limit of forest age for final cutting as specified in Article 38 (v), Article 39 (1), and Article 39 (2) (ii) of the Ordinances of Enforcement of the Forest Act as the criteria for approval of forest management plans.

d) Removals by below-ground biomass until the stand reforested after final cutting reaches the standard cutting age, etc.

$$C_{PJ,RF,BG} = \sum_i C_{PJ,RF,BG,i} = \sum_i (C_{PJ,RF,AG,i} \times R_{ratio,i}) \quad (式6)$$

Symbol	Definition	Unit
$C_{PJ,RF,BG}$	Removals by below-ground biomass until the stand reforested in the relevant fiscal year reaches the standard cutting age, etc.	tCO2
$C_{PJ,RF,BG,i}$	Removals by below-ground biomass until the class i stand reforested in the relevant fiscal year reaches the standard cutting age, etc.	tCO2
$C_{PJ,RF,AG,i}$	Removals by above-ground biomass until the class i stand reforested in the relevant fiscal year reaches the standard cutting age, etc.	tCO2
$R_{ratio,i}$	Factor (underground ratio) for adding removals of the below-ground part (roots) to the removals of the above-ground biomass in class i	—

Supplementary explanation

- $C_{PJ,RF,BG,i}$ shall be limited to an amount equal to the below-ground biomass emissions ($C_{CUT,BG}$ calculated

in Equation 10) from the final cutting implemented prior to reforestation at the relevant reforestation location (if final cutting took place at the relevant forest stand during the certification period and $C_{CUT,BG}$ was not calculated, $C_{PJ,RF,BG,i}$ also cannot be calculated).

e) Removals related to the carbon fixation of harvested timber for the fiscal year that is to be used for a period of time considered to be permanent

$$C_{PJ,WP} = \sum_j C_{PJ,WP,j}$$

$$= \sum_j (C_{SW,C,j} + C_{SW,nC,j} + C_{WB1,C,j} + C_{WB1,nC,j} + C_{WBi,C,j} + C_{WBi,nC,j}) + C_{PW,C}$$

$$+ C_{PW,nC} \qquad \text{（式 7）}$$

Symbol	Definition	Unit
$C_{PJ,WP}$	Removals related to the carbon fixation of harvested timber for the relevant fiscal year that is to be used for a period of time considered to be permanent	tCO2
$C_{PJ,WP,j}$	Removals related to the carbon fixation of species j harvested timber for the relevant fiscal year that is to be used for a period of time considered to be permanent	tCO2
$C_{SW,C,j}$	Removals related to the carbon fixation of species j harvested timber for the relevant fiscal year that is to be used as lumber for building use for a period of time considered to be permanent (See ① below for the calculation method.)	tCO2
$C_{SW,nC,j}$	Removals related to the carbon fixation of species j harvested timber for the relevant fiscal year that is to be use as lumber non-building use for a period of time considered to be permanent (See ② below for the calculation method.)	tCO2
$C_{PW,C}$	Removals related to the carbon fixation of harvested timber for the relevant fiscal year that is to be used as plywood for building use for a period of time considered to be permanent (See ③ below for the calculation method.)	tCO2
$C_{PW,nC}$	Removals related to the carbon fixation of harvested timber for the relevant fiscal year that is to be used as plywood for non-building use for a period of time considered to be permanent (See ④ below for the calculation method.)	tCO2
$C_{WB1,C,j}$	Removals related to the carbon fixation of harvested timber of species j for the relevant fiscal year, for wood chips and wood for lumber converted from wood for raw materials, and wood board produced from wood chips converted from factory offcuts generated in the course of creating wood for plywood as the finished wood product (“wood boards converted from wood for raw materials and factory offcuts”) that is to be used as wood boards for building use for a period of time considered to be permanent (see ⑤ below for the calculation method)	tCO2
$C_{WB1,nC,j}$	Removals related to the carbon fixation of harvested timber of species j for the	tCO2

	relevant fiscal year for wood board converted from wood for raw materials and factory offcuts that is to be used as wood boards for non-building use for a period of time considered to be permanent (see ⑥ below for the calculation method)	
$C_{WBi,C,j}$	Removals related to the carbon fixation of harvested timber of species j for the relevant fiscal year for building-use lumber, and wood board produced from wood chips converted from dismantled plywood ("wood boards converted from dismantled wood") that is to be used as wood boards for building use for a period of time considered to be permanent (see ⑦ below for the calculation method)	tCO2
$C_{WBi,nC,j}$	Removals related to the carbon fixation of harvested timber of species j for the relevant fiscal year for wood board converted from dismantled wood that is to be used as wood boards for non-building use for a period of time considered to be permanent (see ⑧ below for the calculation method)	tCO2
j	Species of harvested timber	—

Supplementary explanation

- Harvested timber includes final cut and thinned timber.
- Wood for lumber, wood for plywood, and wood for raw materials used for wood boards are included in calculations of credits, while wood for raw materials that is to be used for pulp or fuel is not included in calculations.

(1) Removals related to carbon fixation of species j harvested timber for the relevant fiscal year that is to be used as lumber for building use for a period of time considered to be permanent

$$C_{SW,C,j} = RW_{SW,j} \times MY_{SW,j} \times R_{SW,C} \times PY_L \times PS_C \times D \times CC \times \frac{44}{12} \quad (式\ 7-1)$$

Symbol	Definition	Unit
$C_{SW,C,j}$	Removals related to carbon fixation of species j harvested timber for the relevant fiscal year that is to be used as lumber for building use for a period of time considered to be permanent	tCO2
$RW_{SW,j}$	Shipments of species j harvested timber for the relevant fiscal year that is to be used for lumber	m^3
$MY_{SW,j}$	Processing yield at processing from wood from species j for lumber use into lumber	—
$R_{SW,C}$	Ratio of lumber used as lumber for building use	—
PY_L	Processing yield from lumber, plywood, and wood boards used for finished wood products	—
PS_C	Permanency and survival rate for building-use lumber, plywood, and wood boards converted from wood for raw materials and factory offcuts	

D	Wood density (Mass ratio of fully dried wood in relation to volume of air-dried wood)	t/m^3
CC	Wood carbon content (Carbon content of fully dried wood)	—
j	Species of harvested timber	—

(2) Removals related to the carbon fixation of species j harvested timber for the relevant fiscal year that is to be used as lumber for non-building use for a period of time considered to be permanent

$$C_{SW,nC,j} = RW_{SW,j} \times MY_{SW,j} \times R_{SW,nC} \times PY_L \times PS_{SW,nC} \times D \times CC \times \frac{44}{12}$$ （ Formula 7-2）

Symbol	Definition	Unit
$C_{SW,nC,j}$	Removals related to the carbon fixation of species j harvested timber for the relevant fiscal year that is to be used as lumber for non-building use for a period of time considered to be permanent	tCO2
$RW_{SW,j}$	Shipments of species j harvested timber for the relevant fiscal year that is to be used for lumber	m^3
$MY_{SW,j}$	Processing yield at processing from wood from species j for lumber use into lumber	—
$R_{SW,nC}$	Ratio of lumber used as lumber for non-building use	—
PY_L	Processing yield from lumber, plywood, and wood boards used for finished wood products	—
$PS_{SW,nC}$	Permanency and survival rate of lumber for non-building use	—
D	Wood density (Mass ratio of fully dried wood in relation to volume of air-dried wood)	t/m^3
CC	Wood carbon content (Carbon content of fully dried wood)	—
j	Species of harvested timber	—

(3) Removals related to the amount of carbon fixation of harvested timber for the relevant fiscal year that is to be used as plywood for building use for a period of time considered to be permanent

$$C_{PW,C} = RW_{PW} \times MY_{PW} \times R_{PW,C} \times PY_L \times PS_C \times D \times CC \times \frac{44}{12}$$ （式 $7-3$）

Symbol	Definition	Unit
$C_{PW,C}$	Removals related to the amount of carbon fixation of harvested timber for the relevant fiscal year that is to be used as plywood for building use for a period of time considered to be permanent	tCO2
RW_{PW}	Shipments of wood for plywood use from harvested timber for the relevant fiscal	m^3

	year	
MY_{PW}	Processing yield at processing from wood for plywood use to plywood	—
$R_{PW,C}$	Ratio of plywood used as plywood for building use	—
PY_L	Processing yield from lumber, plywood, and wood boards used for finished wood products	—
PS_C	Permanency and survival rate for building-use lumber, plywood, and wood boards converted from wood for raw materials and factory offcuts	—
D	Wood density (Mass ratio of fully dried wood in relation to volume of air-dried wood)	t/m^3
CC	Wood carbon content (Carbon content of fully dried wood)	—

(4) Removals related to the amount of carbon fixation of harvested timber for the relevant fiscal year that is to be used as plywood for non-building use for a period of time considered to be permanent

$$C_{PW,nC} = RW_{PW} \times MY_{PW} \times R_{PW,nC} \times PY_L \times PS_{WBP,nC} \times D \times CC \times \frac{44}{12} \qquad (式\ 7-4)$$

Symbol	Definition	Unit
$C_{PW,nC}$	Removals related to the amount of carbon fixation of harvested timber for the relevant fiscal year that is to be used as plywood for non-building use for a period of time considered to be permanent	tCO2
RW_{PW}	Shipments of wood for plywood use from harvested timber for the relevant fiscal year	m^3
MY_{PW}	Processing yield at processing from wood for plywood use to plywood	—
$R_{PW,nC}$	Ratio of plywood used as plywood for non-building use	—
PY_L	Processing yield from lumber, plywood, and wood boards used for finished wood products	—
$PS_{WBP,nC}$	Permanency and survival rate for non-building use plywood and wood boards converted from wood for raw materials and factory offcuts	—
D	Wood density (Mass ratio of fully dried wood in relation to volume of air-dried wood)	t/m^3
CC	Wood carbon content (Carbon content of fully dried wood)	—

⑤ Removals related to the carbon fixation of harvested timber of species j for the relevant fiscal year for wood board converted from wood for raw materials and factory offcuts that is to be used as wood boards for building use for a period of time considered to be permanent

$$C_{WB1,C,j} = (PR_{WB1,RW} + PR_{WB1,OC,j}) \times R_{WB,C} \times PY_L \times PS_C \times CF_C \times \frac{44}{12}$$

$$PR_{WB1,RW} = RW_{CP} \times CP_{RW} \times WB_{RW}$$

$$PR_{WB1,OC,j} = OC_j \times CP_{OC} \times WB_{OC}$$

$$OC_j = RW_{SW,j} \times (1 - MY_{SW,j}) + RW_{PW} \times (1 - MY_{PW}) + (RW_{SW,j} \times MY_{SW,j} + RW_{PW} \times MY_{PW}) \times (1 - PY_L)$$

（Formula 7-5）

Symbol	Definition	Unit
$C_{WB1,C,j}$	Removals related to the carbon fixation of harvested timber of species j for the relevant fiscal year for wood board converted from wood for raw materials and factory offcuts that is to be used as wood boards for building use for a period of time considered to be permanent	tCO2
$PR_{WB1,RW}$	Volume of wood boards produced from wood chips converted from wood for raw materials from harvested timber for the relevant fiscal year	m^3
$PR_{WB1,OC,j}$	Volume of wood boards produced from wood chips converted from factory offcuts from harvested timber of species j for the relevant fiscal year	m^3
$R_{WB,C}$	Ratio of wood boards used as wood boards for building use	—
PY_L	Processing yield from lumber, plywood, and wood boards used for finished wood products	—
PS_C	Permanency and survival rate for building-use lumber, plywood, and wood boards converted from wood for raw materials and factory offcuts	—
CF_C	Carbon conversion ratio in volume of wood boards for building use	t-C/ m^3
RW_{CP}	Shipments of wood for raw materials from harvested timber for the relevant fiscal year	m^3
CP_{RW}	Ratio of wood chips from raw materials	—
WB_{RW}	Ratio of wood chips converted to wood boards from wood for raw materials	—
OC_j	Volume of factory offcuts generated in the course of creating wood for plywood and wood for lumber as finished products from species j for the relevant fiscal year	m^3
CP_{OC}	Ratio of wood chips from factory cutoffs	—
WB_{OC}	Ratio of wood chips converted to wood boards from factory cutoffs	—
$RW_{SW,j}$	Shipments of species j harvested timber for the relevant fiscal year that is to be used for lumber	m^3
MY_{SW},j	Processing yield at processing from wood from species j for lumber use into lumber	—
RW_{PW}	Shipments of wood for plywood use from harvested timber for the relevant fiscal year	m^3
MY_{PW}	Processing yield at processing from wood for plywood use to plywood	—

⑥ Removals related to the carbon fixation of harvested timber of species j for the relevant fiscal year for wood board converted from wood for raw materials and factory offcuts that is to be used as wood boards for non-building use for a period of time considered to be permanent

$$C_{WB1,nC,j} = (PR_{WB1,RW} + PR_{WB1,OC,j}) \times R_{WB,nC} \times PY_L \times PS_{WBP,nC} \times CF_{nC} \times \frac{44}{12} \quad (式\ 7-6)$$

(For $PR_{WB1,RW}$ and $PR_{WB1,OC,j}$ see ⑤ above)

Symbol	Definition	Unit
$C_{WB1,nC,j}$	Removals related to the carbon fixation of harvested timber of species j for the relevant fiscal year for wood board converted from wood for raw materials and factory offcuts that is to be used as wood boards for non-building use for a period of time considered to be permanent	tCO2
$PR_{WB1,RW}$	Volume of wood boards produced from wood chips converted from wood for raw materials from harvested timber for the relevant fiscal year	m^3
$PR_{WB1,OC,j}$	Volume of wood boards produced from wood chips converted from factory offcuts from harvested timber of species j for the relevant fiscal year	m^3
$R_{WB,nC}$	Ratio of wood boards used as wood boards for non-building use	—
PY_L	Processing yield from lumber, plywood, and wood boards used for finished wood products	—
$PS_{WBP,nC}$	Permanency and survival rate for non-building use plywood and wood boards converted from wood for raw materials and factory offcuts	—
CF_{nC}	Carbon conversion ratio in volume of wood boards for non-building use	t-C/ m^3

⑦ Removals related to the carbon fixation of harvested timber of species j for the relevant fiscal year for wood board converted from dismantled wood that is to be used as wood boards for building use for a period of time considered to be permanent

$$C_{WBi,C,j} = WW_j \times CP_{ww} \times WB_{WW} \times R_{WB,C} \times PY_L \times PS_{WB,C} \times CF_C \times \frac{44}{12}$$

$$WW_j = \left(RW_{SW,j} \times MY_{SW,j} \times R_{SW,C} + RW_{PW} \times MY_{PW} \times R_{PW,C}\right) \times PY_L \times (1 - PS_C) \quad (式\ 7-7)$$

Symbol	Definition	Unit
$C_{WBi,C,j}$	Removals related to the carbon fixation of harvested timber of species j for the relevant fiscal year for wood board converted from dismantled wood that is to be used as wood boards for building use for a period of time considered to be permanent	tCO2
WW_j	Volume of dismantled plywood and wood for lumber used for building use from species j for the relevant fiscal year	m^3

CP_{WW}	Ratio of wood chips from dismantled wood	—
WB_{WW}	Ratio of wood chips converted to wood boards from dismantled wood	—
$R_{WB,C}$	Ratio of wood boards used as wood boards for building use	—
PY_L	Processing yield from lumber, plywood, and wood boards used for finished wood products	—
$PS_{WB,C}$	Permanency and survival rate for wood boards converted from dismantled wood used for building use	—
CF_C	Carbon conversion ratio in volume of wood boards for building use	t-C/ m^3
$RW_{SW,j}$	Shipments of species j harvested timber for the relevant fiscal year that is to be used for lumber	m^3
$MY_{SW,j}$	Processing yield at processing from wood from species j for lumber use into lumber	—
$R_{SW,C}$	Ratio of lumber used as lumber for building use	—
RW_{PW}	Shipments of wood for plywood use from harvested timber for the relevant fiscal year	m^3
MY_{PW}	Processing yield at processing from wood for plywood use to plywood	—
$R_{PW,C}$	Ratio of plywood used as plywood for building use	—
PS_C	Permanency and survival rate for building-use lumber, plywood, and wood boards converted from wood for raw materials and factory offcuts	—

⑧ Removals related to the carbon fixation of harvested timber of species j for the relevant fiscal year for wood board converted from dismantled wood that is to be used as wood boards for non-building use for a period of time considered to be permanent

$$C_{WBi,nC,j} = WW_j \times CP_{ww} \times WB_{WW} \times R_{WB,nC} \times PY_L \times PS_{WB,nC} \times CF_{nC} \times \frac{44}{12}$$ （式 7－8）

(For WW_j see ⑦ above)

Symbol	Definition	Unit
$C_{WBi,nC,j}$	Removals related to the carbon fixation of harvested timber of species j for the relevant fiscal year for wood board converted from dismantled wood that is to be used as wood boards for non-building use for a period of time considered to be permanent	tCO2
WW_j	Volume of dismantled plywood and wood for lumber used for building use from species j for the relevant fiscal year	m^3
CP_{WW}	Ratio of wood chips from dismantled wood	—
WB_{WW}	Ratio of wood chips converted to wood boards from dismantled wood	—
$R_{WB,nC}$	Ratio of wood boards used as wood boards for non-building use	—
PY_L	Processing yield from lumber, plywood, and wood boards used for finished wood	—

	products	
$PS_{WB,nC}$	Permanency and survival rate for wood boards converted from dismantled wood used for non-building use	—
CF_{nC}	Carbon conversion ratio in volume of wood boards for non-building use	t-C/ m^3

4. **Calculation of emissions after project implementation (emissions from final cutting)**

Emissions in the class in which the final cutting took place are recorded for in the fiscal year of final cutting.

（Formula 8）

Symbol	Definition	Unit
C_{cut}	Removals after project implementation in the fiscal year	tCO2
$C_{cut,AG}$	Emissions of above-ground biomass associated with final cutting in the relevant fiscal year	tCO2
$C_{cut,BG}$	Emissions of below-ground biomass associated with final cutting in the relevant fiscal year	tCO2

a） Emissions of above-ground biomass associated with final cutting

（Formula 9）

Symbol	Definition	Unit
$C_{cut,AG}$	Emissions of above-ground biomass associated with final cutting in the relevant fiscal year	tCO2
$C_{cut,AG,i}$	Emissions of above-ground biomass in class i due to final cutting in the relevant fiscal year	tCO2
$Area_{Forest,cut,i}$	Area of the forest stand classified as class i where forest management (final cutting) is implemented in the relevant fiscal year	ha
$Trunk_{SC,cut,i}$	Stem volume per unit area in class i before final cutting	m^3/ha
WD_i	Factor (volume density) for converting stem volume growth in class i to biomass (dry weight)	t/m^3
BEF_i	Factor (expansion factor) for adding branch and leaf biomass to stem biomass in class i	—
CF	Factor for converting biomass amount (dry weight) to carbon amount (Carbon content)	—
i	Class of the forest by forest type (artificial forest, natural regenerated forest),	—

	tree species, forest age, site index, etc.	

Supplementary explanation

- Final cutting includes clearcutting, selective cutting, and other cutting done for renewal or preparation for renewal.
- In the $Area_{Forest,cut,i} \times Trunk_{SC,cut,i}$ part of Equation 9, the volume of standing timber harvested (m^3) as indicated in the logging notice under Article 15 of the Forest Act (Act No. 249 of 1951) may be substituted.
- If a decision has been made for land conversion of the project site for purposes such as public roads or power lines, or there have occurred natural disturbances or final cutting for preventing damages by disease, insects or animals, which is based on law or regulation documented by the central or local government, and these decision or occurrence were due to unavoidable reasons such as unforeseeability at the time of development of PDD, the area of the relevant locations will not be included in $Area_{Forest,i}$ and $Area_{Forest,cut,i}$ thereafter. However, if the locations meet the definition of the methodology (see 3. and 4.) after the locations ares restored to forest, they shall be included in $Area_{Forest,i}$ and $Area_{Forest,cut,i}$.
- And if final cutting is planned or implemented as a measure against the above disease/insect/animal damages, evidence materials such as policies, guidelines, plans, guidance notices, etc. issued by the national or local government at the time of registration/certification must be submitted to Management.

b) Emissions of below-ground biomass associated with final cutting

(Formula 10)

Symbol	Definition	Unit
$C_{cut,BG}$	Emissions of below-ground biomass associated with final cutting in the relevant fiscal year	tCO2
$C_{cut,BG,i}$	Emissions of below-ground biomass in class i due to final cutting in the relevant fiscal year	tCO2
$C_{cut,AG,i}$	Emissions of above-ground biomass in class i due to final cutting in the relevant fiscal year	tCO2
$R_{ratio,i}$	Factor for adding below-ground (root) biomass emissions to above-ground biomass emissions in class iUnderground ratio	—
i	Class of the forest by forest type (artificial forest, natural regenerated forest), tree species, forest age, site index, etc.	—

5. Baseline removals approach

The baseline removals in this methodology are the removals that would have occurred if appropriate forest management or forest protection based on the forest management plan had not been implemented from the start of the certification period until the relevant fiscal year.

（Formula 11）

Symbol	Definition	Unit
C_{BL}	Baseline removals for the relevant fiscal year	tCO2

6. Monitoring methods

The table below lists monitoring items and examples of monitoring methods required to calculate removals and emissions after project implementation. When developing a PDD, monitoring items are determined according to the selected calculation formulas, and a monitoring plan is drafted according to Implementation Rule (for project participants) and Monitoring and Calculation Rules. Monitoring is done according to the monitoring plan.

1) Monitoring of activity

Monitoring items		Monitoring methods	Monitoring frequency	Notes
$Area_{Forest,i}$	Of the artificial forests in class i where forest management was carried out from April 1990 to the relevant fiscal year, the area (ha) of forest stands where forest management (excluding final cutting) or forest protection was implemented from the start of the certification period until the relevant fiscal year, and the area (ha) of naturally regenerated forests in class i where forest protection was implemented (only areas designated as restricted forests, such as forest reserves, etc.) from the start of the certification period until the relevant fiscal year	If artificial forest: ・ (Area where forest management was implemented) Measurements by compass surveying, etc. However, the results of past measurements conducted for grant applications, etc. may be used. ・ (Status of forest management) Confirmation of forest management records, logging notices, photos that can be used to determine traces and timings of previous work in the relevant forest stand, etc. ・ (Status of forest protection) Confirmation of records such as work logs If natural regenerated forest: ・ (Area where forest protection was implemented) Measurements by compass surveying, etc. for extermination and prevention of forest pests. However, the results of past measurements conducted for grant applications, etc. may be used.	Once at the time of application for verification However, except in special circumstances, the value at the time of initial verification application may be used for area of forest management or protection	*1

		Area of forest register for forest sub-compartments included in zone for fire prevention and other protection. ・ (Status of forest protection) Confirmation of work logs, photos of the state of forests monitored, etc.		
$Area_{Forest,RF,i}$	Area of the forest stand classified as class i where forest management is implemented in the relevant fiscal year after final cutting (ha)	・ Actual measurement by compass surveying, etc. (existing actual measurement results may be used)	Once at the time of reforestation	*1
$RW_{SW,j}$	Shipments of species j harvested timber for the relevant fiscal year that is to be used for lumber (m^3)	・In principle, proof is by purchase slips related to shipments of logs by use (for lumber, plywood, and raw materials) and the transaction history of CoC certified timber. However, if the recipient of the shipment sorts the unprocessed timber and the sender is unable to obtain data on the shipments by use itself, the calculation is done as follows. ① If statistics on shipments of local lumber by use in the prefecture to which the project site belongs (including biomass fuel) are available, determine the ratio of wood for lumber, plywood, and raw materials (total amount of timber for pulp, wood chips, and fuel) to the total log shipments based on statistics published in the fiscal year before the fiscal year in which the certification request was made, and multiply them by the total log shipments from the project for which the breakdown by use is not specified to calculate the proportionate amounts. ② If ① above is not feasible, calculate the percentage of timber	Totaled over applicable period	*2
RW_{PW}	Shipments of wood for plywood use from harvested timber for the relevant fiscal year（m^3）			
RW_{CP}	Shipments of wood for raw materials from harvested timber for the relevant fiscal year (m^3)			

		used for lumber, plywood, and wood for raw materials (total amount of timber for pulp, wood chips, and fuel) from the total amount produced domestically, based on the most recently published Ministry of Agriculture, Forestry and Fisheries "Wood Supply and Demand Chart", and multiply the total unprocessed timber shipments from the project for which the breakdown by use is not specified to calculate the proportionate amount.		
$Area_{Forest,cut,i}$	Area of the forest stand classified as class i where forest management (final cutting) is implemented in the relevant fiscal year (ha)	・ Actual measurement by compass surveying, etc. (existing actual measurement results may be used)	Once at the time of final cutting	*1

2) Factor monitoring

Monitoring items		Monitoring methods	Monitoring frequency	Notes
$Trunk_{SC,i}$	Annual stem volume growth per unit area for the fiscal year in class i (m^3/ha). For natural regenerated forest, the following additional items will be monitored. (1) Stem growth per unit are by forest age class (m^3/ha) (2) Stem growth per unit by forest age class obtained from the results of the Basic Survey of Forest Ecosystem Diversity (m^3/ha)	・ In principle, use values from yield estimate tables prepared by prefectures (including, deciduous stand harvest table, standard stock table, etc.) that are appropriate for the forests covered by the project ・ For stem growth per unit are by forest age class in natural regenerated forests based on the Basic Survey of Forest Ecosystem Diversity, use values provided from the Forestry Agency	Once at the time of application for verification	*3
$Trunk_{SC,RF,i}$	Stem volume per unit of area in reforested class i stand standard cutting age, etc. (m^3/ha)	・ In principle, use values from yield estimate tables prepared by prefectures that are appropriate for the forests covered by the project	Once at the time of application for verification	*3
$Trunk_{SC,cut,i}$	Stem volume per unit area in class i before final cutting (m^3/ha)	・ In principle, use values from yield estimate tables prepared by	Once at the time of application for	*3

		prefectures that are appropriate for the forests covered by the project	verification	
WD_i	Factor (volume density) for converting stem volume growth in class i to biomass (dry weight)	・ In principle, use values published in the National Greenhouse Gas Inventory Reports of JAPAN and other sources that are appropriate to the characteristics of the forests	[Request frequency] Use the latest at the time of request for verification	*4
BEF_i	Factor (expansion factor) for adding branch and leaf biomass to stem biomass in class i			
CF	Factor for converting biomass amount (dry weight) to carbon amount (Carbon content)			
$R_{ratio,i}$	Factor for adding below-ground (root) CO_2 removals (emissions) to above-ground biomass CO_2 removals (emissions) in class iUnderground ratio			
i	Class of the forest by forest type (artificial forest, natural regenerated forest), tree species, forest age, site index, etc.	・For artificial forest (except young forests for which no tree height is given in the yield estimate table), measure diameter at chest height and tree height for site index monitoring ・For artificial forest (except young forests for which no tree height is given in the yield estimate table), in principle, read the site index applied to the species of the stand from the forest registry and apply ・For natural regenerated forest, site index monitoring is not required.	Once at the time of initial verification application (However, for artificial forests, if after identifying the site index in a young forest where tree height is not listed in the harvest estimate table, application for verification is from the time the forest reaches the age for which tree height is indicated in the harvest estimate table, monitoring must be conducted again.)	*5
$MY_{SW,j}$	Processing yield at processing from wood from species j for lumber use into lumber	・ In principle, use log conversion rates of the Ministry of Agriculture, Forestry and Fisheries “Wood Supply and Demand Chart”	Use materials published in the fiscal year before the fiscal year in which the certification request was made	*6
MY_{PW}	Processing yield at processing from wood for plywood use to plywood			

CP_{RW}	Ratio of wood chips from wood for raw materials・Set to 1.	・Set to 1	－	*7
CP_{OC}	Ratio of wood chips from factory cutoffs ・Set to 0.501.	・Set to 0.501	－	*7
CP_{WW}	Ratio of wood chips from dismantled wood・Set to 0.898.	・Set to 0.898	－	*7
WB_{RW}	Ratio of wood chips converted to wood boards from wood for raw materials	・Set to 0.012	－	*8
WB_{OC}	Ratio of wood chips converted to wood boards from factory cutoffs	・Set to 0.087.	－	*8
WB_{WW}	Ratio of wood chips converted to wood boards from dismantled wood・Set to 0.121.	・Set to 0.121	－	*8
$R_{SW,C}$	Ratio of lumber used as lumber for building use	・In principle, the ratio of lumber for building use (industrial lumber) and lumber for non-building use (civil engineering lumber, wooden box construction boards, packaging materials, furniture and fittings materials, other) out of shipment of lumber by use (domestic) in the Ministry of Agriculture, Forestry and Fisheries “Wood Supply and Demand Report” published in the fiscal year before the fiscal year in which the certification request was made shall be used.	Use materials published in the fiscal year before the fiscal year in which the certification request was made	*6
$R_{SW,nC}$	Ratio of lumber used as lumber for non-building use			
$R_{PW,C}$	Ratio of plywood used as plywood for building use	In principle, the ratio by use calculated from statistical values in the Ministry of Agriculture, Forestry and Fisheries “Wood Supply and Demand Report” published in the fiscal year before the fiscal year in which the certification request was made shall be used. Specifically, calculate the ratio by use, with (a) and (b) as for building use and (c) and (d)	Use materials published in the fiscal year before the fiscal year in which the certification request was made	*6
$R_{PW,nC}$	Ratio of plywood used as plywood for non-building use			

		as for non-building use, as shown below. (a) Production volume of conifers for structural use out of regular plywood production multiplied by the percentage of domestic wood in plywood veneer consumption (b) Production volume of domestic wood in structural use out of LVL production (c) Production volume regular plywood other than (a) multiplied by the percentage of domestic wood in plywood veneer consumption (d) Production volume of domestic wood in others (other than structural use) out of LVL production		
$R_{WB,\ C}$	Ratio of wood boards used as wood boards for building use	・Set to 0.758.	―	*7
$R_{WB,\ nC}$	Ratio of wood boards used as wood boards for non-building use・Set to 0.242.	・Set to 0.242	―	*7
PY_L	Processing yield from lumber, plywood, and wood boards used for finished wood products	・In principle, set as 0.9.	―	*6
PS_C	Permanency and survival rate for building-use lumber, plywood, and wood boards converted from wood for raw materials and factory offcuts	・Set as 0.167. (90-year survival rate estimated based on floor area data and section residual rate estimation method from the Ministry of Internal Affairs and Communications' "Summary Survey of Fixed Asset Prices, etc.")	―	*7
$PS_{WB,C}$	Permanency and survival rate for wood boards converted from dismantled wood used for building use・Set to 0.736. ―	・Set to 0.736		*7
$PS_{SW,nC}$	Permanency and survival rate of lumber for non-building use	・Set as 0.170. (90-year survival rate for the first-order decay function	―	*7

		model for lumber as presented in the IPCC guidelines)		
$PS_{WBP,nC}$	Permanency and survival rate for non-building use plywood and wood boards converted from wood for raw materials and factory offcuts	・Set as 0.084. (90-year survival rate for the first-order decay function model for wood panel as presented in the IPCC guidelines)	－	*7
$PS_{WB,nC}$	Permanency and survival rate for wood boards converted from dismantled wood used for non-building use	・Set to 0.417.	－	*7
D	For wood density (mass ratio of fully dried wood in relation to volume of air-dried wood in t/m^3)	and lumber, in principle, Attachment 3 of Monitoring and Calculation Rule (for forest management projects) shall be used. ・For plywood, in principle, set as 0.542. (Values shown in National Greenhouse Gas Inventory Report of JAPAN)	－	*9
CC	Wood carbon content (Carbon content of air-dried wood)	・In principle, set as 0.5 for lumber and 0.493 for plywood. (Values shown in National Greenhouse Gas Inventory Report of JAPAN)	－	*9
CF_C	Carbon conversion ratio in volume of wood boards for building use (t-C/ m^3)	・ Set to 0.252.	－	*9
CF_{nC}	Carbon conversion ratio in volume of wood boards for non-building use (t-C/ m^3)	・Set 0.205.	－	*9

＜*1＞

- Forestry roads and forest operation roads are excluded from the forest area. However, this does not apply to forestry roads if they are included in the forest area in the forest management plan prepared in accordance with the operations of each prefecture.
- Refer to 2.3 to 2.3.3 of Monitoring and Calculation Rule (for forest management projects) concerning monitoring of the area of the forest.
- Photos of the forest to be monitored shall be taken to verify the status of the forest when forest management or protection are implemented, and the photo taken shall be stored by the project

participant. Refer to 2.4.1 of Monitoring and Calculation Rule (for forest management projects) for details. However, where forest protection has been implemented for forests with an age of 11 years or older through aircraft patrols or other measures, photographs shall be taken to show the state of the forest canopy in the forests being monitored.

＜*2＞

- Refer to 2.8.1 of Monitoring and Calculation Rule (for forest management projects) concerning monitoring of shipment of harvested timber.

＜*3＞

- Refer to 2.5 to 2.5.3 of Monitoring and Calculation Rule (for forest management projects) for monitoring of stem volume growth and stem volume.
- The Local Yield Table Construction System (LYCS) or other literature/documentation may be used in addition to the yield tables and yield estimate tables prepared by prefectures, but in the case of citing stem volume growth rates from documentation that have not been checked by a third party (reviewers of academic papers, etc.), the validity of the relevant reference documentation must be confirmed by a validation authority.
  - Forestry and Forest Products Research Institute Website: http://www2.ffpri.affrc.go.jp/labs/LYCS/index.html
- In the case of planting with “elite trees” or other saplings that grow faster than normal forest trees, and when there is no yield estimate table etc. appropriate for the forest at the project site, the stem volume growth may be measured. Refer to 2.5.1.4 of Monitoring and Calculation Rule (for forest management projects) for the measurement method.

＜*4＞

- Refer to 2.6 of Monitoring and Calculation Rule (for forest management projects) for monitoring of bulk density, expansion factor, underground ratio and carbon content.
- Other literature/documentation may be used in addition to the National Greenhouse Gas Inventory Report of JAPAN, but if bulk density, expansion factor, underground ratio, and carbon content are cited from a document that has not been checked by a third party (academic paper reviewer, etc.), the validity of the supporting documents must be confirmed by a validation authority.

＜*5＞

- Refer to 2.7.5 of Monitoring and Calculation Rule (for forest management projects) for site index monitoring. Refer to 2.7.4 of the same rule for identification of the site index of young forests for which no tree height is given in the yield estimate table and to 2.7.5 of the same rule for site index identification when calculating removals until the stand reforested after final cutting reaches the standard cutting age, etc.

＜*6＞

- Refer to 2.8.2、2.8.5 and 2.8.6 of Monitoring and Calculation Rule (for forest management projects) for monitoring of processing yield for lumber and plywood, processing yield for finished wood products, and ratio of lumber and plywood by use (building or non-building use)

- In addition to the values noted in the monitoring method, when self-measured values can be reported, the measured values can be used.

＜*7＞

- For monitoring the ratio of wood chips by raw material, the ratio of wood boards by use (building or non-building use), permanency and survival rate values by product/by use, and the carbon conversion ratio of wood boards, refer to 2.8.3, 2.8.6, 2.8.7, and 2.8.10 of Monitoring and Calculation Rule (for forest management projects) for details.

＜*8＞

- For the ratio of wood chips converted to wood boards by raw material, refer to 2.8.4 of Monitoring and Calculation Rule (for forest management projects) for details.
- For the ratio of wood chips from raw materials converted into wood boards, in addition the values shown in the monitoring method, actual measurement values may be used when self-measured values can be reported.

＜*9＞

- Refer to 2.8.8 and 2.8.9 of Monitoring and Calculation Rule (for forest management projects) for monitoring of wood density and wood carbon content.
- In addition to the values noted in the monitoring method, other literature/documentation may be used, but if wood density and wood carbon content are quoted from a document that has not been checked by a third party (e.g., academic paper reviewer), the validity of the supporting documents must be confirmed by the validation authority.
-

7. Addendum

1) Evaluation of additionality

- When activities applying this methodology fall under any of the following, an evaluation of additionality is not required due to the high probability of having economic barriers.
  - Final cutting based on the forest management plan is not planned in the project site during the certification period
  - Reforestation based on the forest management plan is planned in all project sites where final cutting based on the forest management plan is planned during the certification period

  If none of the above apply, additionality shall be determined based on the following criteria

  Profits at the project site in the certification period < expenses at the project site in the certification period (expenses required for forest management - grants + interest on bank and other loans) + expenses up to the date when 10 years has elapsed from the ending date of the certification period for stands where final cutting was done during the certification period (expenses required for reforestation and nurturing - grants + interest on bank and other loans)

*Profits from forest management for the forest at the project site, expenses required for forest management, etc. shall be proved by using documents that enable the actual income and expenditures to be identified. However, in the absence of forest-specific data for the project site, statistical data from the national and prefectural governments and other government agencies (forest management expenses, mountain base standing tree prices, etc.), program software to predict forest management income and expenditure, etc., may be used to calculate profits and expenses when standard forest management is implemented.

2) Certification period

- The ending date of the certification period for projects to which this methodology is applied shall be the ending date of the fiscal year that includes the date when 8 years have elapsed from the starting date of the certification period or the ending date of an arbitrarily selected fiscal year in the period from year from the date when 9 years have elapsed to the date when 16 years have passed. However, for reforestation in stands where final cutting was conducted during the certification period, removals can be calculated for reforestation conducted from the final date of the certification period to the date when 2 years have elapsed, and certification for such reforestation can be applied for until the ending date of the fiscal year following that in which such reforestation was conducted.
- When applying for certification of removals described in the proviso above, certification of the removals during (part of) the certification period may also be applied for at the same time as a special exception.

[Explanation]

- Special exception to application for removals during (part of) the certification period is applicable when applying together for (1) removals of stands other than final cutting stands for which application for certification could not be made during the certification period because the emissions of final cutting stands resulted in negative removals for the entire project site for the relevant fiscal year during the certification period and for (2) removals of the portion for which certification was not applied during the certification period when applying for certification of removals equivalent to the standard cutting age, etc. of the reforested stands by the day on which 2 years have elapsed since the end of the certification period.

3) Registration of programmatic projects

- Programmatic projects cannot be registered under this methodology.

4) Photography, etc. during the reforestation monitoring period

- When applying for certification of credits claiming the amount of carbon stock corresponding to the standard cutting age, etc. for a forest stand where reforestation was implemented after regeneration cutting as removal volume of the relevant project under 3 of this methodology, photos that show the current status of the relevant forest stand must be submitted to the system manager each year during the period until the forest age for the relevant forest stand becomes 10 and every other year during the period from the time when the forest age is 12 until the expiration of the reforestation monitoring period, by June 30 of the following year. In such instance, this may be replaced with submission of aerial photos by drone, etc. or satellite images by which it can be

visually confirmed that the relevant forest stand is not affected by cutting, development or natural disturbance, etc. judging by interpreting the tree crowns.

[Explanation]

Photographs on the ground shall be taken in the following manner.

① If forest age is 10 years or less, in the forest to be monitored, photos are taken that show the status of planting, weeding, etc. of the relevant forest stand.

② If forest age is 11 years or greater, in the forest to be monitored, one photo is taken showing the state of the forest interior and forest floor, and another photo from the same direction, either horizontally or diagonally upward, showing the state of the forest canopy.

③ Photographs are taken using a digital camera, and information on the date, time, and location of each photograph is also stored.

<List of documents that need to be prepared for validation>

Required documents	Specific examples
Documents showing that eligibility criteria are satisfied	・Forest management plan and corresponding certificates, etc. that include the project site for which project plan is registered (plan for management [reforestation, nurturing, or thinning] to satisfy Eligibility Criterion 3 is stated in the forest management plan) ・Forest plan diagrams, etc., that includes the forest to be registered in the project plan. ・If a forest owner who does not participate in the forest plan that includes the forest to be registered in the project plan exists, evidence of confirmation and agreement with project participants on keeping permanence of the forest ・If the owner of the forest where final cutting is planned in the forest management plan at the time of plan application is unable to participate in the project, a statement of reasons for that Acknowledgement indicating intention to continue the forest management plan during the certification period and up to the date when 10 years have elapsed from the end of the certification period and indicating understanding of duty of replenishment if certification of the forest management plan is not continued

<Details regarding the formulation and revision of methodologies>

Ver	Date of enactment or revision	Expiration date	Details
1.0	2013.5.10	2014 7 20	Newly enacted
2.0	2014.1.20	2014.12.25	● For Criterion 3 of eligibility criteria, changed to “the cumulative amount of expected removals during the certification period is positive” ● Added “nurturing” to forest management practices ● Added “status of forest management practices or protection” to the monitoring items

			● Specified that the area where planting, nurturing, and thinning activities have been conducted shall be calculated by multiplying the measured area by 0.9 ● Changed to "Exclude the location in question from the project site" when unavoidable land conversion is done
2.1	2014.12.26	2017.3.13	● Clarified what must satisfy eligibility criteria
2.2	2017.3.14	2017.7.25	● Changed the description of CF (carbon content) ● Changed the reference for some coefficients from Report on Japan's Supplementary Information on LULUCF activities under Article 3, Paragraphs 3 and 4 of the Kyoto Protocol to National Greenhouse Gas Inventory Report of JAPAN
2.3	2017.7.26	2019.2.27	● Added "<Definition of Terms>" ● Removed the term, "forest management practices plan" ● Modified Eligibility Criterion 2 ● Modified the definition, supplementary explanation, monitoring method, and monitoring frequency for "$Area_{Forest,i}$" ● Removed the description of $M_{Forest,i}$ ● Modified the definition of "　$Trunk_{SC,i}$" ● Modified the definition of "i" ● Revised the description of baseline removals approach
2.4	2019.2.28	2021.8.2	● Modified supplementary explanation of calculation formula for removals by above-ground biomass (formula 3) ● Modified supplementary explanation of emissions of above-ground biomass associated with final cutting (formula 6) ● Modified the definition of symbol "i" in the calculation formulas ● Added when emissions after project implementation (emissions from final cutting) do not need to be considered
2.5	2021.8.3	2021.12.1	● Added an addendum to check forest status at time of forest protection ● Added supplementary explanation on the definition of final cutting and noted that the product of the area of final cutting and area of stem volume can be substituted for the volume of standing timber harvested described in the logging notice in the calculation of emissions
2.6	2021.12.2	2022.8.9	● Added Eligibility Criterion 3 to the cases where planting or nurturing plans can be used instead of thinning ● Revised description regarding photography in the addendum

			● Modified list of documents that need to be prepared for validation
3.0	2022.8.10	2022.12.18	● Adjustments made to definitions of terms ● Modified eligibility criteria and added Criterion 5 ● Added removals of reforested stands and removals related to carbon fixation of harvested timber to removals after project implementation ● Added the area of natural regenerated forests where forest protection was implemented to the area subject to calculation of removals ● Added to monitoring methods content related to removals of reforested stands, removals related to carbon fixation of harvested timber, and removals of natural regenerated forests where forest protection was implemented ● Revised in the addendum rules related to determination of additionality, certification period, programmatic projects, and reforestation monitoring period
3.1	2022.12.19	2023.3.1	● Modified supplementary explanation related to “Removals by biomass until the stand reforested in the relevant fiscal year reaches the standard cutting age, etc.” in Formula 2 ● Added note on yield estimate table in monitoring method of “stem growth volume” monitoring item ● Modified formula related to profits and expenses for evaluating additionality ● Modified rules related to the ending date of the certification period
3.2	2023.3.2	—	● Revised the requirements for Eligibility Criteria in cases where the project participant registers a project plan by selecting a portion of the forest owned or managed by the project participant itself, and added examples for “non-arbitrarily selected” Added descriptions of the calculation of removals related to the carbon fixation of wood boards